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1. Introduction

In today's digitally connected world, a secure and efficient network infrastructure is vital to support daily operations and long-term growth. Haile Hotels and Resorts is constructing a new hotel facility in Debre Birhan Regio-Politan City, employing over 500 staff members across various departments. To support its operations and communication needs, the hotel requires a well-structured, scalable, and secure network.

This mini-project presents a comprehensive logical network design tailored for Haile Hotels and Resorts, focusing on reliability, security, and performance. The proposed solution follows a hierarchical network model that separates the core, distribution, and access layers for optimal traffic flow and management. Technologies such as VLAN segmentation, inter-VLAN routing, dynamic IP address allocation via DHCP, secure remote access via SSH, and wireless access points for specific areas are incorporated.

The network is designed and simulated using Cisco Packet Tracer and includes routing with OSPF, centralized DHCP services, and secure configurations to support the hotel's departmental needs and ensure future scalability. Special considerations have also been taken to restrict administrative remote access to authorized personnel only, maintaining a strong security posture.

This documentation outlines the overall design objectives, technology stack, and implementation approach to meet the business and technical requirements of the hotel.

2. Scope of the Project

The scope of this mini-project is to design, simulate, and document a complete logical network solution for Haile Hotels and Resorts' newly constructed facility in Debre Birhan Regio-Politan City. The network aims to support over 500 users distributed across three floors and several departments, each requiring secure, isolated, and reliable connectivity.

The design is developed and tested using **Cisco Packet Tracer version 8.2.2** and is limited to simulation only. The project follows a hierarchical network model and implements modern enterprise networking best practices to ensure future scalability and operational efficiency.

In-Scope Features

The network design and simulation include the following components and configurations:

- **Inter-VLAN Routing:** Implemented on multilayer core switches to allow communication between departmental VLANs.
- **IP Addressing Scheme:**
 - Static IP configuration for servers in the Server Room.
 - DHCP-based dynamic addressing for all other devices.
- **OSPF Routing Protocol:** Used for dynamic route advertisement across routers and multilayer switches.
- **NAT (Network Address Translation):** Deployed on the edge router to convert private IP addresses to a public IP address for internet access through the ISP.
- **Wireless Network:** Configured specifically for the Restaurant Department using WRT300N wireless routers.

- **SSH Configuration:** Secured remote access enabled only for authorized administrative users; access is strictly limited to designated devices.
- **Server Room Setup:** Contains the DHCP server and other critical systems with static IPs. This server handles IP address allocation across all departments.
- **Topology Components:**
 - 1 ISP Router
 - 1 Edge Router
 - 2 Core Switches (Layer 3)
 - 3 Distribution Switches
 - 7 Access Switches
 - End devices including generic PCs, WRT300N wireless routers, and servers.

User and Departmental Coverage

The network serves over **500 users** across **three floors**, with each department assigned to a **unique VLAN and subnet**. The segmentation provides performance benefits and enhances security through logical isolation.

Project Constraints

- The implementation is limited to **simulation only** and does not include physical deployment.
- The design and testing environment is restricted to **Cisco Packet Tracer version 8.2.2**.
- Hardware used in the simulation (routers, switches, PCs, servers) is virtual and based on Cisco Packet Tracer's available device models.

Project Deliverables

- **Network Documentation:** A comprehensive report detailing the design, addressing, configurations, and testing procedures (this document in PDF).
- **Simulation File:** A Cisco Packet Tracer project file (.pkt) containing the fully implemented and configured network design.

3. Logical Network Design

The proposed network for Haile Hotels and Resorts is structured using a **three-tier hierarchical architecture**, which organizes the network into Core, Distribution, and Access layers. This model promotes scalability, performance, and easier troubleshooting by separating roles and responsibilities across layers. The entire network is simulated using **Cisco Packet Tracer** and designed to meet both current and future operational demands.

3.1 Core Layer

The core layer consists of **two multilayer Layer 3 switches** that provide high-speed switching and routing between VLANs. These switches are configured for **inter-VLAN routing** and form the backbone of the network, enabling efficient communication between departments. Redundancy is ensured by linking both core switches to the distribution switches, preventing single points of failure and enhancing fault tolerance.

3.2 Distribution Layer

The distribution layer is composed of **three switches**, each connected redundantly to both core switches. This layer aggregates traffic from the access layer and enforces **policies and route filtering** between VLANs and floors. It acts as a mediator between the high-speed core and department-focused access layer, handling forwarding decisions and path optimization.

3.3 Access Layer

At the access layer, **seven switches** are deployed to serve different departments across the hotel's three floors. These switches connect directly to end-user devices and ensure reliable communication within each department. Each access switch is mapped to a specific department or section:

- **First Floor**
 - Sales and Marketing Department – 58 users
 - Human Resource and Logistics Department – 60 users
- **Second Floor**
 - Restaurant – 150 users (also includes wireless coverage)
 - Finance and Accounts Department – 50 users
 - Administrator and Public Relations Department – 40 users
- **Third Floor**
 - ICT Department – 30 users
 - Server Room – 12 devices (servers, DHCP, etc.)

All access switches are uplinked to the distribution switches, ensuring organized traffic flow and access control.

3.4 Wireless Network Integration

The **Restaurant Department** includes a **wireless network**, implemented using the **WRT300N** wireless router in Cisco Packet Tracer. This provides mobility and flexibility for restaurant staff and devices while maintaining secure integration into the network.

3.5 Internet Connectivity and NAT

An **ISP Router** provides internet access to the hotel, and an internal **Border Router** is configured with **Network Address Translation (NAT)** to convert private IP addresses into a public IP for communication with the ISP. The border router is connected to both core switches for redundancy and internet traffic distribution.

3.6 Server Room and DHCP

The **Server Room** is connected to an access switch on the third floor and houses critical infrastructure including the **DHCP Server**, which dynamically allocates IP addresses to all client devices across departments and VLANs. All server room devices are assigned static IP addresses for stability and consistent configuration.

4. IP Addressing Scheme

To ensure logical separation, efficient routing, and easier management, each department at Haile Hotels and Resorts is assigned its own VLAN and IP subnet. A centralized DHCP server located in the Server

Room dynamically assigns IP addresses to all end-user devices across the network. Critical devices in the Server Room use static IP configurations.

The IP addressing plan uses private Class C networks (192.168.X.0) with subnet sizes matched to the number of users in each department. Default gateways are assigned on .1 of each subnet, and DHCP pools begin from .2 onward.

4.1 VLAN and IP Allocation Table

VLAN ID	Department	Subnet	Subnet Mask	Default Gateway	DHCP Range Start	Max Users
10	Sales & Marketing	192.168.10.0/26	255.255.255.192	192.168.10.1	192.168.10.2	58
20	HR & Logistics	192.168.20.0/26	255.255.255.192	192.168.20.1	192.168.20.2	60
30	Restaurant	192.168.30.0/24	255.255.255.0	192.168.30.1	192.168.30.2	150
40	Finance & Accounts	192.168.40.0/26	255.255.255.192	192.168.40.1	192.168.40.2	50
50	Admin & Public Relations	192.168.50.0/26	255.255.255.192	192.168.50.1	192.168.50.2	40
60	ICT Department	192.168.60.0/27	255.255.255.224	192.168.60.1	192.168.60.2	30
70	Server Room	192.168.70.0/28	255.255.255.240	192.168.70.1	192.168.70.10 Static IPs-only	12

4.2 DHCP Configuration

- **Centralized DHCP Server** in the Server Room.
- Each VLAN has its own DHCP pool.
- Gateway IP .1 is excluded and manually configured on Layer 3 switches.
- DNS Server used: 8.8.8.8 (Google DNS).
- Server Room devices use static IPs for reliability and service availability.

5. VLAN Configuration Plan

To enhance security, improve traffic management, and logically segment departments, each floor and department in Haile Hotels and Resorts is assigned a unique **VLAN**. This configuration ensures that broadcast traffic is limited within departments, and access control policies can be enforced with greater precision.

5.1 VLAN Structure

Each VLAN corresponds to a specific department. This setup allows better management of departmental resources, isolates traffic between departments, and supports inter-VLAN communication via multilayer switches.

VLAN ID	Department	Floor	No. of Users	Purpose/Notes
10	Sales & Marketing	First Floor	58	Departmental devices and staff
20	HR & Logistics	First Floor	60	Includes personnel and logistics
30	Restaurant	Second Floor	150	Includes wireless access
40	Finance & Accounts	Second Floor	50	Sensitive financial systems
50	Admin & Public Relations	Second Floor	40	Admin coordination and communication
60	ICT Department	Third Floor	30	Technical staff and IT support
70	Server Room	Third Floor	12	Servers and DHCP (static IPs)

5.2 VLAN Implementation

- VLANs are created on **Layer 3 Core Switches**.
- VLAN interfaces (SVIs) are configured with .1 addresses (e.g., 192.168.10.1) for routing.
- All **Access Switch ports** are assigned to corresponding VLANs based on floor and department.
- Wireless router for the Restaurant is connected to an Access Switch port configured as part of VLAN 30.
- Inter-VLAN routing is enabled via Core Switches using SVIs.
- Trunk links are used between:
 - Access Switches ↔ Distribution Switches
 - Distribution Switches ↔ Core Switches
 - VLANs are allowed on trunk ports using switchport trunk allowed vlan command.

6. Routing and NAT Configuration

To ensure efficient communication between VLANs and to provide internet access to all departments, the network design implements both **OSPF (Open Shortest Path First)** as a dynamic routing protocol and **NAT (Network Address Translation)** at the network edge.

6.1 Inter-VLAN Routing with Layer 3 Switches

Inter-VLAN routing is enabled through **Switched Virtual Interfaces (SVIs)** configured on the two **Core Layer 3 Switches**. These SVIs serve as default gateways for each VLAN and allow hosts in separate VLANs to communicate.

- Each VLAN has its own SVI (e.g., `interface vlan 10` with IP 192.168.10.1)
- SVIs are enabled with routing capability via the `ip routing` command
- Routing occurs internally on the Core Switches without the need to pass traffic to routers

6.2 OSPF Configuration

The **Open Shortest Path First (OSPF)** protocol is used for dynamic route exchange between:

- The **Border Router** and **Core Switches**
- The **Core Switches** and **Distribution Switches**, if necessary

This ensures that:

- All VLAN subnets are advertised and reachable
- Changes in topology (e.g., link failure) are quickly adapted to
- Future scalability is supported through area-based structure

Sample OSPF Commands (on a Core Switch):

```
router ospf 1
network 192.168.10.0 0.0.0.63 area 0
network 192.168.20.0 0.0.0.63 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.63 area 0
network 192.168.50.0 0.0.0.63 area 0
network 192.168.60.0 0.0.0.31 area 0
network 192.168.70.0 0.0.0.15 area 0
```

6.3 NAT Configuration on Border Router

Since the internal network uses **private IP addresses**, **NAT (specifically PAT)** is configured on the **Border Router** to translate internal IPs into a single public IP address before traffic reaches the ISP router.

Sample NAT Configuration:

```
interface GigabitEthernet0/0/1
ip address 20.0.0.1 255.255.255.252
ip nat inside

interface GigabitEthernet0/0/0
ip address 10.0.0.1 255.255.255.252
ip nat inside

interface Serial0/1/0
ip address 200.200.200.2 255.255.255.252
ip nat outside

ip nat inside source list 1 interface g0/1 overload
```

- **NAT inside** is configured on the internal-facing interface

- **NAT outside** is configured on the ISP-facing interface
- **Overload (PAT)** allows multiple private hosts to share a single public IP

6.4 Redundancy and Failover Considerations

- The **Border Router** is connected to both **Core Switches** for reliability
- OSPF convergence and load balancing are enabled by default
- **Static routes** are not used to avoid rigidity and manual maintenance

7. Network Security and Access Control

Security is one of the most critical aspects of any enterprise-grade network, especially for large organizations like Haile Hotels and Resorts. A well-designed security plan ensures that sensitive data is protected, access is restricted to authorized users, and operations continue smoothly without disruption.

The following security mechanisms are implemented in the proposed network design:

7.1 VLAN-Based Segmentation and Isolation

All departments are logically segmented using **Virtual LANs (VLANs)**. Each VLAN represents a department and has its own subnet, isolating traffic and reducing the risk of unauthorized access or broadcast storms. This design enhances both security and performance by ensuring:

- Departmental **broadcast domains** are contained.
- Communication between departments occurs only when allowed, through **Layer 3 inter-VLAN routing**.
- Each department's data is logically isolated from others, making unauthorized lateral movement across departments more difficult.

7.2 Secure Remote Access with SSH

To protect the network infrastructure from unauthorized access, all routers and Layer 3 switches are configured to use **Secure Shell (SSH)** instead of Telnet. SSH ensures that management traffic is encrypted, preventing attackers from intercepting sensitive information such as usernames and passwords.

The following steps were taken to implement secure SSH access:

```
ip domain-name hailehotels.com

crypto key generate rsa

username admin secret Password123

line vty 0 4
  login local
  transport input ssh
exit
```

This ensures encrypted remote access and enforces **local authentication** using a defined username and secure password.

7.3 Restricting SSH Access Using Access Control Lists (ACLs)

To further tighten security, SSH access is limited **only to authorized devices** from the **Admin & Public Relations Department (VLAN 50)**.

The network uses an **Access Control List (ACL)** to enforce this restriction:

```
ip access-list standard SSH-ONLY-ADMIN
  permit 192.168.50.0 0.0.0.63
  deny any
exit

line vty 0 4
  access-class SSH-ONLY-ADMIN in
  login local
  transport input ssh
exit
```

This configuration allows only the administrative staff from VLAN 50 to initiate SSH sessions, while denying all other access attempts from other subnets.

By implementing this policy, the network upholds the **Principle of Least Privilege**, ensuring that only authorized users have management access to critical devices.

8. Network Testing and Verification

Once the network topology was implemented in Cisco Packet Tracer, a series of comprehensive tests were performed to validate the functionality, performance, and security of the entire system. These tests confirmed that all design requirements were met and that the system could operate reliably under expected conditions.

8.1 IP Addressing Verification

To ensure correct IP address allocation:

- **DHCP Lease Validation:** Devices in all VLANs were tested to confirm that they received IP addresses from the centralized DHCP server in the Server Room.
- **Static IP Devices:** Devices in the Server Room (e.g., DHCP server, core switches) were assigned static IP addresses, and proper connectivity was confirmed.
- **Gateway Reachability:** Each client was able to ping its default gateway (e.g., 192.168.10.1 for VLAN 10).

✓ Result: All devices received correct IP configurations, and gateway reachability was successful.

8.2 Inter-VLAN Routing Test

To verify routing between VLANs:

- Devices from different VLANs (e.g., VLAN 10 and VLAN 40) were tested using the `ping` command.
- Inter-VLAN communication was validated through Layer 3 core switches.

✓ Result: Devices were able to communicate across VLANs as intended, confirming that **SVIs and routing** were functioning correctly.

8.3 Internet Access via NAT

To confirm internet connectivity:

- Internal clients initiated pings and simulated web access to external destinations.
- The Border Router was verified to successfully **translate private IPs to a public IP** using NAT.

✓ Result: NAT on the Border Router translated traffic appropriately, and internet access (via ISP simulation) was successful.

8.4 Wireless Connectivity

The wireless network for the Restaurant Department was tested as follows:

- Laptops connected to the **WRT300N router** using the configured SSID.
- Each wireless device successfully obtained an IP from the DHCP server.
- Wireless clients were able to ping other internal departments and access simulated external resources.

✓ Result: Wireless communication was functional, with full network and internet access.

8.5 SSH Access Control Test

To test secure remote access:

- SSH sessions were initiated from an **authorized device in VLAN 50** (Admin & PR).
- Attempts were made from unauthorized VLANs (e.g., VLAN 10 or VLAN 60) and were denied by the ACL.
- Correct username (`admin`) and password (`Password123`) were required.

✓ Result: SSH was **only accessible from VLAN 50**, as per ACL configuration, proving that security controls were properly enforced.

8.6 DHCP Server Test

The centralized DHCP server was tested for each scope:

- Devices in each VLAN successfully received addresses from their correct pools.
- The DHCP server respected excluded addresses and assigned IPs starting from .2.
- Leases were monitored and validated in the DHCP configuration panel.

✓ Result: All DHCP pools worked correctly with no IP conflicts.

8.7 Trunk and VLAN Tagging Verification

- All trunk ports were checked to ensure correct VLAN tagging and allowed VLANs.
- Switches forwarded tagged traffic between access and distribution layers without errors.

✓ Result: VLANs were propagated across the network correctly.

8.8 Overall System Integrity

The full network was tested for:

- End-to-end connectivity between any two clients
- DHCP lease renewals and router uptime
- OSPF route visibility (`show ip route ospf`)
- Core switch routing tables and neighbor adjacency

✓ Result: Network stability, redundancy, and functionality were confirmed.

9. Conclusion

The network design presented in this project addresses the current and anticipated future needs of Haile Hotels and Resorts' newly constructed facility in Debre Birhan Regio-Politan City. With over 500 staff members distributed across three floors and multiple departments, the organization requires a secure, scalable, and high-performance network infrastructure — all of which have been successfully delivered through this logical design.

By adopting a **three-tier hierarchical architecture**, the network ensures efficient traffic flow, clear separation of responsibilities, and simplified management. Each department has been assigned a **dedicated VLAN** and subnet, supported by **inter-VLAN routing** configured on multilayer switches. The use of **OSPF** as the routing protocol guarantees dynamic and reliable route propagation, while **NAT** on the Border Router enables secure internet access for internal devices using private IP addresses.

Security has been tightly integrated into the network:

- All remote administrative access is protected via **SSH** with encrypted sessions.
- An **ACL** ensures only authorized administrative devices (VLAN 50) can access critical infrastructure.
- Password protection, VLAN segmentation, and a centralized DHCP service further enhance the network's integrity.

Wireless connectivity in the Restaurant Department extends network accessibility while maintaining isolation and security through VLAN integration. The **centralized DHCP server**

streamlines IP address management for all departments, while **static addressing** ensures reliability for essential services like DNS and DHCP.

The entire network has been simulated and verified in **Cisco Packet Tracer**, confirming that all devices can:

- Obtain correct IP addresses,
- Communicate across VLANs where permitted,
- Access the internet through NAT,
- And be remotely managed securely where authorized.

In conclusion, this logical network design meets the technical requirements of the hotel, ensures business continuity, supports future expansion, and follows enterprise networking best practices. The solution is both robust and adaptable, ready to scale as the organization grows or as new technologies are introduced.